

## KE-04: Strict interval occupancy across block Lanczos iterations

Proposed resolution for mathematical review

**Status of this manuscript:** Proposed resolution, not independently verified.

**Outcome claimed:** Affirmative resolution claim.

**Prepared:** 11 September 2026.

**Target:** [Repository entry KE-04](#), snapshot b412319.

This is a proposed argument generated in a ChatGPT conversation and prepared for mathematical review. No independent referee report or formal proof certificate accompanies it. Supporting algebraic and numerical diagnostics are not a substitute for proof review. No publication or priority claim is made.

### Theorem KE-04: proposed strict interval occupancy

Let  $A \in \mathbb{R}^{n \times n}$  be symmetric and let  $V \in \mathbb{R}^{n \times p}$  have full column rank. Define

$$\mathcal{K}_\ell = \text{range}[V, AV, \dots, A^{\ell-1}V].$$

Assume  $\dim \mathcal{K}_s = sp$  and therefore full block dimension at all preceding iterations. Let  $P_\ell$  be the orthogonal projector onto  $\mathcal{K}_\ell$ , and let  $H_\ell = P_\ell A|_{\mathcal{K}_\ell}$ . Write its eigenvalues, with multiplicity, as

$$\theta_1^{(\ell)} \leq \dots \leq \theta_{\ell p}^{(\ell)}.$$

These are the Ritz values and are independent of the chosen orthonormal basis.

**Claim.** For every  $1 \leq k < j \leq s$  and every  $1 \leq i \leq (k-1)p$ , the open interval

$$(\theta_i^{(k)}, \theta_{i+p}^{(k)})$$

contains an eigenvalue of  $H_j$ .

#### 1. Assume a gap at the later iteration

There are no admissible indices when  $k = 1$ , so suppose  $k \geq 2$ . Put

$$a = \theta_i^{(k)}, \quad b = \theta_{i+p}^{(k)}, \quad q(t) = (t-a)(t-b).$$

Suppose that  $H_j$  has no eigenvalue in  $(a, b)$ . Since  $a \leq b$ ,

$$q(H_j) \succeq 0.$$

Let  $E \subseteq \mathcal{K}_k$  be the span of an orthonormal set of eigenvectors of  $H_k$  corresponding to indices  $i, \dots, i+p$ . Then

$$\dim E = p+1, \quad \langle x, q(H_k)x \rangle \leq 0 \quad (x \in E).$$

The codimension of  $\mathcal{K}_{k-1}$  in  $\mathcal{K}_k$  is  $p$ , so  $E \cap \mathcal{K}_{k-1}$  contains a nonzero vector  $x$ .

#### 2. The quadratic forms agree on that vector

Since  $x \in \mathcal{K}_{k-1}$ , we have  $Ax \in \mathcal{K}_k$  and

$$H_k x = H_j x = Ax.$$

Self-adjointness of the compressions gives

$$\begin{aligned} \langle x, q(H_j)x \rangle &= \|Ax\|^2 - (a+b)\langle x, Ax \rangle + ab\|x\|^2 \\ &= \langle x, q(H_k)x \rangle \leq 0. \end{aligned}$$

Combined with  $q(H_j) \succeq 0$ , this implies  $q(H_j)x = 0$ .

Moreover  $A^2x \in \mathcal{K}_{k+1} \subseteq \mathcal{K}_j$ , so  $H_j^2x = A^2x$ . Hence

$$q(A)x = 0.$$

### 3. Full block rank excludes this annihilation

Write

$$x = \sum_{\ell=0}^{k-2} A^\ell V c_\ell.$$

Let  $r$  be the largest index with  $c_r \neq 0$ ; such an index exists because  $x \neq 0$ . The polynomial  $q$  is monic of degree two. Therefore the coefficient of  $A^{r+2}V$  in the resulting block-power expansion of  $q(A)x$  is exactly  $c_r \neq 0$ .

Since  $k+1 \leq j \leq s$ , the block Krylov matrix

$$[V, AV, \dots, A^k V]$$

has full column rank. As  $r+2 \leq k$ , the nonzero coefficient just found implies  $q(A)x \neq 0$ , a contradiction.

This establishes the claim for all admissible indices and iterations. If  $a = b$ , the same argument applies with  $q(t) = (t - a)^2$  and rules out that case; thus the proof does not presuppose strictly separated endpoints or simple Ritz values.  $\square$

### Scope and review notes

The argument uses real symmetric  $A$ , a full-column-rank starting block, exact arithmetic, and full block dimension through iteration  $j$ . It includes eigenvalue multiplicities and the possibility of coincident proposed interval endpoints, which it rules out in the relevant range. It does not assert a finite-precision analogue.

The key review point is the equality of the two quadratic forms on a vector in  $\mathcal{K}_{k-1}$ , followed by conversion to a degree-two annihilating polynomial. Full rank is required through  $\mathcal{K}_{k+1}$ , which is guaranteed by  $k+1 \leq j \leq s$ .

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

### Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry KE-04](#), repository snapshot b412319, accessed 11 September 2026.

D. Šimonová and P. Tichý, *On finite precision block Lanczos computations*, [arXiv manuscript, version 1](#), 22 July 2025, §2.1, unnumbered conjecture. The catalog identifies this as the source of its exact-arithmetic interval statement.